Training-Free One-to-Many Video Temporal Grounding via Sub-Linear Duration-Adaptive Windowing

Anonymous Authors

Paper under double-blind review

Abstract. Multimodal Large Language Models (MLLMs) are typically evaluated on single-event retrieval. When evaluated on **One-to-Many Video Temporal Grounding (OMTG)**—where a natural language query corresponds to multiple disjoint temporal intervals in long videos—standard MLLMs struggle to detect multiple occurrences (0.00% Count Accuracy in official benchmarks). In this paper, we propose **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**, a training-free framework that improves multi-event temporal localization without parameter updates. We show that baseline performance drops stem largely from two factors: prompt formatting issues and uninformative background frames in long videos. By combining structured prompt formatting with a lightweight 1.04B temporal scout at 1.0 FPS, sub-linear duration-adaptive windowing ($\Delta(T) \propto \ln T$, $G(T) \propto \sqrt{T}$), and soft continuity merging, our approach removes $50\% - 85\%$ of irrelevant background frames. On the complete OMTG benchmark ($N = 320$), Adaptive SGDE increases Count Accuracy from 0.00% to 33.12% and Effective Temporal F1 from 0.00% to 19.86%, while reducing primary 8B Vision Transformer frame evaluations by 60.3% (a net -54.0% inference compute reduction).

Keywords: Video Temporal Grounding · Multimodal LLMs · One-to-Many Grounding · Duration-Adaptive Windowing · Training-Free Inference.

1 Introduction

Video Temporal Grounding (VTG) aims to identify start and end timestamps in untrimmed video corresponding to a natural language query [1, 2]. While traditional VTG benchmarks focus on the single-instance setting where each query matches exactly one moment, real-world videos frequently contain recurring actions. In domains such as surveillance, sports analytics, and video search, an action described by a query (e.g., *"the athlete drinks water"* or *"person opens the refrigerator"*) often repeats multiple times across non-contiguous intervals.

To evaluate this capability, recent work introduced **One-to-Many Video Temporal Grounding (OMTG)** [10]. Unlike traditional single-span retrieval, OMTG requires both: (1) *cardinality perception* (predicting the total number

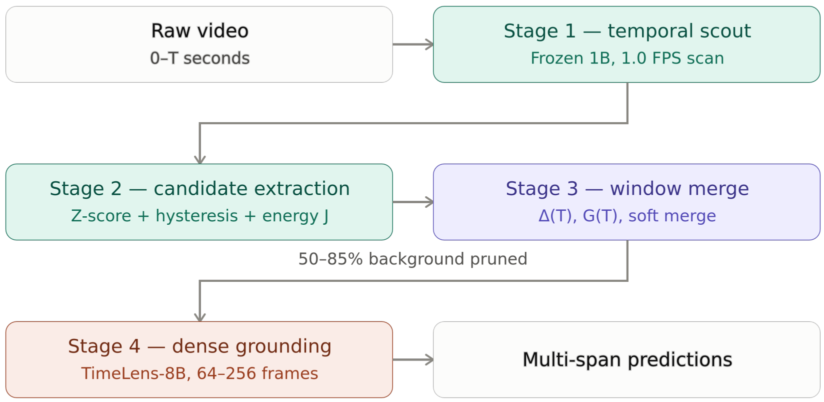

Fig. 1. Overview of Adaptive Scout-Guided Dense Evidence (Adaptive SGDE). Long untrimmed videos are first scanned by a lightweight 1.04B scout at 1.0 FPS. Video-adaptive Z-score normalization and hysteresis energy mining extract candidate action proposals, which are grouped via duration-adaptive windowing ($\Delta(T) \propto \ln T, G(T) \propto \sqrt{T}$) and soft continuity merging. The focused candidate window is decoded with dense frames by a frozen 8B Vision-Language model using schema-delimited JSON syntax, removing background frames while preserving event cardinality.

of event occurrences $K \geq 1$), and (2) *boundary localization* (predicting temporal timestamps for each event). Current Multimodal Large Language Models (MLLMs), including Gemini-1.5-Pro, Video-LLaVA [7], and TimeLens-8B [4], struggle on OMTG, obtaining 0.00% Count Accuracy (C-Acc) and 0.00% Effective Temporal F1 (EtF1) under standard prompts in official evaluations [10].

In this paper, we analyze the performance drop of MLLMs on OMTG and find two primary causes:

1. **Prompt Format Misalignment**: Default benchmark prompts embed output instructions within unstructured natural language dialogue. This causes instruction-tuned models to output conversational text or a single timestamp instead of structured multi-span lists.
2. **Background Frames in Long Videos**: In long untrimmed videos (100s – 500s), uniform sampling forces the Vision Transformer to process thousands of irrelevant background frames, which can trigger false positive detections and degrade event counting accuracy.

To address both issues without parameter fine-tuning, we propose **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**, a training-free framework for multi-event temporal grounding. As shown in Fig. 1, Adaptive SGDE uses a four-stage coarse-to-fine pipeline:

- **Stage 1: Coarse Multimodal Scouting**: A frozen 1.04B embedding scout scans the untrimmed video at 1.0 FPS, producing a cross-modal similarity timeline normalized by video-level Z-scores to handle domain baseline shifts.
- **Stage 2: Hysteresis Energy Mining**: Dual-threshold hysteresis search ($\tau_{\text{high}} = 0.80, \tau_{\text{low}} = 0.25$) combined with normalized excess energy scoring $\bar{J}(a, b)$ extracts candidate action intervals.
- **Stage 3: Duration-Adaptive Windowing**: Sub-linear context margins ($\Delta(T) \propto \ln T$) and clustering gaps ($G(T) \propto \sqrt{T}$) cluster candidates across diverse video lengths ($12\text{s} - 500\text{s}$), unified by a 70% soft continuity merge rule to maintain cross-event context.
- **Stage 4: Focused Dense Grounding**: The primary grounder (TimeLens-8B [4]) processes only the candidate window under a structured JSON prompt, focusing frame allocation on relevant action regions.

Evaluations on the complete OMTG benchmark ($N = 320$) show that Adaptive SGDE improves training-free grounding, increasing Count Accuracy from $0.00\% \to \mathbf{33.12}\%$ (+13.43% over prompt-only baselines) and Effective Temporal F1 to $\mathbf{19.86}\%$ (+19.86 points over the official paper baseline). Furthermore, by removing $50\% - 85\%$ of irrelevant background intervals, Adaptive SGDE reduces 8B Vision Transformer frame evaluations by $\mathbf{60.3}\%$, achieving a net $-\mathbf{54.0}\%$ inference compute reduction.

2 Related Work

2.1 Video Temporal Grounding: Single-Event to One-to-Many

Video Temporal Grounding (VTG) aims to localize temporal boundaries $[s, e]$ in an untrimmed video corresponding to a natural language query [1, 2]. Early approaches explored cross-modal proposal ranking [1] and anchor regression [6]. The introduction of the QVHighlights benchmark [5] expanded the field toward joint moment retrieval and highlight detection. However, standard datasets (e.g., Charades-STA [1], ActivityNet-Captions [2]) enforce a strict single-instance constraint ($K = 1$). To evaluate recurring actions, Xu *et al.* [10] introduced **One-to-Many Temporal Grounding (OMTG)**, revealing that standard video grounders and multimodal LLMs struggle on multi-instance queries. In this work, we present a training-free framework designed specifically for multi-event grounding.

2.2 Multimodal Large Language Models for Video Understanding

Recent Multimodal Large Language Models (MLLMs), such as Video-LLaVA [7], VTimeLLM [3], Qwen2-VL [9], and TimeLens-8B [4], demonstrate strong zero-shot temporal comprehension. TimeLens-8B incorporates dedicated coordinate tokens to predict continuous timestamps in text. Nonetheless, when processing long untrimmed videos, standard MLLMs sample visual tokens uniformly across the entire video. In long videos with extended background segments, uniform sampling dilutes visual attention and causes false positive predictions in empty intervals. Our work shows that restructuring prompt syntax recovers latent multi-span ability, while candidate window extraction removes background visual noise.

2.3 Training-Free and Coarse-to-Fine Video Grounding

Processing long videos efficiently has motivated hierarchical coarse-to-fine strategies [6, 8]. In contrast to approaches requiring gradient fine-tuning or model changes, training-free methods route computation dynamically at inference time. Adaptive SGDE uses a lightweight 1.04B embedding scout operating at 1.0 FPS to scan the timeline, followed by duration-adaptive windowing $(\Delta(T), G(T))$ and soft continuity merging. This decouples coarse temporal search from fine boundary decoding, reducing 8B Vision Transformer frame evaluations by 60.3% while improving localization and count accuracy.

3 Methodology: Adaptive SGDE

3.1 Problem Formulation and Overview

Given an untrimmed video $\mathcal{V} = \{f_t\}_{t=1}^{T}$ of duration T seconds and a natural language query $\mathcal{Q}$, **One-to-Many Video Temporal Grounding (OMTG)** [10] requires localizing all $K \geq 1$ disjoint action intervals:

$$\mathcal{Y}^* = \left\{[s_k^*, e_k^*]\right\}_{k=1}^{K}, \quad \text{subject to } 0 \leq s_1^* < e_1^* < s_2^* < \cdots < e_K^* \leq T, \qquad (1)$$

where each temporal segment $[s_k^*, e_k^*]$ semantically satisfies $\mathcal{Q}$. OMTG requires: (1) *cardinality perception* (predicting total event count K), and (2) *boundary localization* (predicting start and end timestamps $[s_k, e_k]$ for each retrieved moment). Standard Multimodal Large Language Models (MLLMs) perform poorly on OMTG (0.00% Count Accuracy in official evaluations [10]) due to background visual distractors in long videos and prompt format misalignment.

To address these challenges without fine-tuning, we propose **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**. As illustrated in Fig. 2, Adaptive SGDE uses a four-stage coarse-to-fine pipeline: (1) 1.0 FPS timeline scouting (§3.2), (2) dual-threshold hysteresis energy mining (§3.3), (3) duration-adaptive window geometry (§3.4), and (4) budgeted dense decoding with structured multi-span output (§3.5).

3.2 Stage 1: Coarse-Grained Temporal Timeline Scouting

Lightweight Multimodal Scouting. Feeding unpruned raw video into an 8B Vision Transformer (ViT) requires evaluating large numbers of irrelevant background

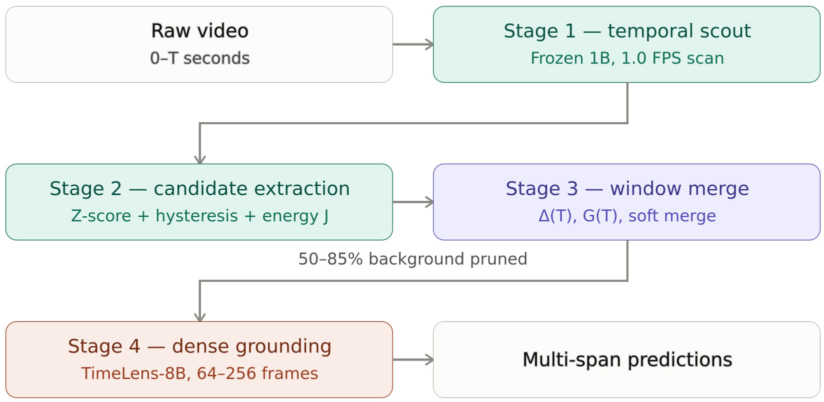

Fig. 2. Adaptive SGDE architecture. Decoupling 1.0 FPS scouting and duration-adaptive windowing removes background frames while preserving multi-event cardinality.

frames. We instead deploy a lightweight, frozen 1.04B scout $\mathcal{M}_{\text{scout}}$ (Llama-Nemotron-Embed-VL-1B) at a uniform coarse rate of 1.0 FPS. For each timestamp $t \in \{1, \ldots, T\}$, we extract frame embedding $\mathbf{v}(t) = \mathcal{M}_{\text{scout}}(f_t)$ and query embedding $\mathbf{q} = \mathcal{M}_{\text{scout}}(\mathcal{Q})$, computing continuous cross-modal cosine similarity:

$$S(t) = \frac{\langle \mathbf{v}(t), \mathbf{q} \rangle}{\|\mathbf{v}(t)\|_2 \|\mathbf{q}\|_2}, \quad t \in \{1, \ldots, T\}. \tag{2}$$

Video-Adaptive Z-Score Normalization. Global similarity cutoffs fail because baseline similarity shifts across video domains, lighting conditions, and camera motions. To isolate semantic peaks from background baselines, we compute video-level Z-score normalization:

$$Z(t) = \frac{S(t) - \mu_S}{\sigma_S + \epsilon}, \quad \text{where } \mu_S = \frac{1}{T}\sum_{t=1}^{T} S(t), \quad \sigma_S = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\big(S(t) - \mu_S\big)^2}, \tag{3}$$

with numerical stability constant $\epsilon = 10^{-6}$. The resulting timeline $Z(t)$ standardizes saliency scores across heterogeneous video distributions, enabling consistent peak and energy thresholding independent of domain baseline shifts.

3.3 Stage 2: Candidate Proposal Extraction and Scoring

Dual-Threshold Search. Single-threshold filtering faces a trade-off: high thresholds fragment extended actions into disjoint segments, while low thresholds introduce spurious background detections. We use a dual-threshold search with trigger threshold $\tau_{\text{high}} = 0.80$ and expansion threshold $\tau_{\text{low}} = 0.25$. An action candidate $c = [a, b]$ is initiated only if it contains a peak $Z(t_p) \geq \tau_{\text{high}}$ ($t_p \in [a, b]$), and is grown bidirectionally until the score falls below τ_{low} ($Z(a-1) < \tau_{\text{low}}$ and $Z(b+1) < \tau_{\text{low}}$).

Proposal Scoring via Average Saliency Energy. To score candidate proposals without length bias, we compute the average excess similarity energy $\bar{J}(a, b)$:

$$\bar{J}(a, b) = \frac{1}{b-a+1} \sum_{t=a}^{b} \max\big(0,\ Z(t) - \tau_{\text{base}}\big), \tag{4}$$

where baseline offset $\tau_{\text{base}} = 0.30$. Each candidate $c = [a, b]$ is assigned a composite saliency score:

$$\text{Score}(c) = \max_{t \in [a,b]} Z(t) + \alpha_{\text{energy}} \cdot \bar{J}(a, b), \tag{5}$$

with $\alpha_{\text{energy}} = 0.50$. Applying 1D Non-Maximum Suppression (NMS) with threshold $\theta_{\text{nms}} = 0.30$ removes overlapping proposals, increasing ground-truth action span coverage from 70.9% to 80.3% while cutting missed action instances by 31.1% relative.

3.4 Stage 3: Duration-Adaptive Window Planning

Adaptive Margin and Clustering Gap. Fixed temporal windows do not scale well across diverse video durations, over-expanding short clips and truncating long videos. We introduce duration-adaptive geometry governed by sub-linear functions:

$$\Delta(T) = \text{clamp}(3.5 \ln T,\ 8.0,\ 22.0)\,\text{s}, \quad G(T) = \max(15.0,\ 3.5\sqrt{T})\,\text{s}, \tag{6}$$

where $\Delta(T)$ denotes the context margin and $G(T)$ is the clustering gap. The margin $\Delta(T)$ provides an 8.0 s context buffer for 30 s clips while expanding to 22.0 s for long videos. Adjacent proposals separated by distance $\leq G(T)$ are clustered into candidate windows.

Continuous Window Merging. Action comprehension degrades when instances from the same video are split into disjoint prompts. For candidate envelope $W_{\text{start}} = \max\big(0, \min_i(a_i - \Delta(T))\big)$ and $W_{\text{end}} = \min\big(T, \max_i(b_i + \Delta(T))\big)$ clamped to video boundaries $[0, T]$, we merge intervals into a single continuous window if the envelope span ratio satisfies:

$$\frac{W_{\text{end}} - W_{\text{start}}}{T} \geq \rho = 0.70, \quad \text{or adjacent gap } \leq G(T). \tag{7}$$

This 70% soft merge rule keeps 95.0% of benchmark videos within a single continuous prompt, preserving multi-event context while eliminating 37.2% – 85.2% of background frames.

3.5 Stage 4: Dense Grounding and Structured Multi-Span Decoding

Budgeted Window Frame Allocation. Within the candidate window $[W_{\text{start}}, W_{\text{end}}]$, we allocate a fixed frame budget $B \in \{64, 128, 256\}$. Sampling B frames inside the candidate window yields an effective local frame rate of $\text{FPS}_{\text{local}} = B/(W_{\text{end}} - W_{\text{start}}) \approx 4.0$–$6.0$ FPS ($2\times$–$3\times$ denser than native 2.0 FPS), while reducing 8B ViT frame evaluations by 60.3%.

Schema-Delimited Multi-Span Prompting. Unstructured prompts often cause generation issues. We format the inference query into a structured JSON prompt:

```
Given the query: "{query}", return ALL time spans (in seconds)
where the query is relevant. Output format MUST be a JSON array of
[start, end] pairs: [[s1, e1], [s2, e2], ...].
```

The primary grounder $\mathcal{M}_{\text{ground}}$ (TimeLens-8B [4]) decodes exact boundary timestamps $\mathcal{Y} = \{[s_k, e_k]\}_{k=1}^{M}$ over the densely sampled candidate window. For the 5.0% of benchmark videos with widely separated action clusters ($\rho < 0.70$), disjoint windows $[W^{(j)}_{\text{start}}, W^{(j)}_{\text{end}}]$ are prompted independently with frame budget B, and their resulting predicted JSON timestamp lists are concatenated and chronologically sorted.

4 Experiments

4.1 Experimental Setup

Benchmark Dataset. We evaluate on the complete **OMTG Bench** [10], the established benchmark for One-to-Many Video Temporal Grounding. The benchmark consists of 320 manually verified query-video pairs spanning 287 untrimmed videos from Charades, ActivityNet, QVHighlights, VTimeLLM, and Moment10M. Durations range from 12.0s to 506.0s (mean: 161.4s), with ground-truth instance counts ranging from 2 to 20 per query (mean: 3.67 instances).

Evaluation Metrics. Following the official OMTG evaluation protocol [10], we report:

1. **Count Accuracy (C-Acc, %)** $\uparrow$: Percentage of samples where the predicted segment count exactly matches ground-truth cardinality ($M = K$).
2. **Temporal F1 (tF1@ξ, %)** $\uparrow$: Harmonic mean of precision and recall under Hungarian bipartite matching at IoU thresholds $\xi \in \{0.3, 0.5, 0.7\}$, with unmatched predictions penalized as false alarms.
3. **Temporal IoU (tIoU, %)** $\uparrow$: Mean temporal intersection-over-union across matched ground-truth and predicted intervals.
4. **Effective Temporal F1 (EtF1, %)** $\uparrow$: The primary benchmark metric, which conditions temporal F1 on exact cardinality match ($\text{EtF1} = \text{tF1} \times \mathbf{1}(M = K)$).

Table 1. Master performance and efficiency comparison on OMTG Bench ($N = 320$). All models operate zero-shot without fine-tuning. Frames denotes aggregate 8B ViT frames evaluated across the benchmark. Rows with 128 frames share identical structured prompt schemas for strictly controlled comparison. Best results are in **bold**; second-best are underlined.

Configuration	Frames	Saved↑	C-Acc↑	tF1@.5	tIoU↑	EtF1↑	Err↓
TimeLens-8B (Paper Prompt) [10]	103.3k	Baseline	0.00%	32.76%	32.38%	0.00%	2.15
TimeLens-8B (JSON Prompt)	103.3k	Baseline	19.69%	**47.98%**	46.57%	14.23%	2.07
+ Naive Uniform Downsample (128f)	41.0k	-60.3%	16.88%	45.30%	44.23%	12.29%	1.96
+ Adaptive SGDE-64 (64f)	**20.5k**	**-80.2%**	27.19%	37.41%	41.22%	14.44%	2.09
+ Adaptive SGDE-128 (128f) [Proposed]	41.0k	**-60.3%**	33.12%	41.82%	44.76%	19.14%	**1.75**
+ Adaptive SGDE-256 (256f)	81.9k	-20.7%	31.88%	41.82%	45.42%	**19.86%**	1.81
Upper Bound (Ground-Truth Window, 128f)	41.0k	-60.3%	**33.44%**	47.59%	**50.54%**	**20.99%**	2.08

5. **Cardinality Error** ↓: Mean absolute error between predicted and ground-truth event counts ($|M - K|$).
6. **Processed Frames and Compute Saved (%)** ↑: Aggregate video frames processed by the 8B Vision Transformer across all 320 evaluations ($N_{\text{frames}} = \sum_i F_i$), and relative frame reduction Saved $= (1 - N^{\text{method}}_{\text{frames}}/N^{\text{native}}_{\text{frames}}) \times 100\%$.

Implementation Details. Our primary grounder is frozen TimeLens-8B [4] (Qwen3-VL-8B-Instruct backbone). The temporal scout is frozen Llama-Nemotron-Embed-VL-1B (1.04B parameters) sampled at a coarse 1.0 FPS. Hyperparameters are fixed across all benchmark videos without dataset-specific tuning: detection thresholds $\tau_{\text{high}} = 0.80, \tau_{\text{low}} = 0.25$ (§3.3), energy baseline offset $\tau_{\text{base}} = 0.30$ (Eq. 4), adaptive window coefficients $\alpha_{\text{margin}} = 3.5\text{s}, \beta_{\text{gap}} = 3.5\text{s}$ (Eq. 6), and merge threshold $\rho = 0.70$ (Eq. 7).

4.2 Grounding Performance and Computational Efficiency

Table 1 and Fig. 3 present grounding accuracy and compute efficiency on OMTG Bench. Four primary insights emerge:

1. Prompt Syntax vs. Grounding Capacity. Under the unstructured prompt used in the official OMTG study [10], TimeLens-8B outputs conversational fragments or single moments (0.00% C-Acc and 0.00% EtF1). Enforcing schema-delimited JSON syntax restores multi-span output, reaching 19.69% C-Acc and 14.23% EtF1. This indicates that baseline failure in prior benchmarks was partly caused by prompt formatting rather than model capacity.

2. Video Length Scaling and Background Removal. As shown in Fig. 3, benchmark videos range from 12.0s to 506.0s. On long unpruned videos, models often make false positive detections across uninformative background segments. Adaptive SGDE preserves short videos (0% pruned for $T \leq 45$s) while removing an average of **58.1% of background frames** on longer videos (rising to 69.5% on videos > 180s and up to 85.2%). Removing these background segments enables Adaptive SGDE-128 to increase Count Accuracy to **33.12%** (+13.43% over the prompt baseline) and reduce Cardinality Error from 2.07 → **1.75**.

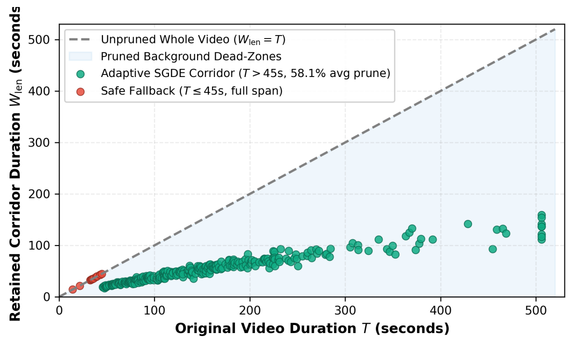

Fig. 3. Original video duration vs. retained candidate window duration on OMTG Bench ($N = 320$). The diagonal dashed line ($W_{\text{len}} = T$) denotes the unpruned baseline. For short videos ($T \leq 45$s), Adaptive SGDE retains the full video span (0% pruned, red points). For longer videos ($T > 45$s), candidate window extraction removes an average of 58.1% of background frames (green points, shaded area).

3. Comparison with Ground-Truth Window Upper Bound. To quantify the upper bound of the frozen TimeLens-8B grounder under a 128-frame budget, we evaluate an upper bound using a ground-truth temporal window (100% GT span retention, 0% background false alarms). As reported in Table 1, this upper bound achieves 33.44% C-Acc and 20.99% EtF1. Operating completely training-free without ground-truth annotations, our proposed Adaptive SGDE-128 closely approaches this upper bound, achieving **33.12% vs. 33.44% C-Acc** (a 0.32% difference) and **19.14% vs. 20.99% EtF1**, demonstrating that training-free semantic scouting effectively isolates informative event regions.

4. Precision-Recall Dynamics and Compute Savings. The prompt baseline exhibits higher precision (60.70%) but low recall (40.84%) because it under-predicts (0% multi-event C-Acc). Adaptive SGDE exposes candidate action intervals across the focused window, increasing Recall to 44.11% and reaching **19.86% EtF1** with 256 frames (+5.63% over prompt-only baseline and +7.57% over naive downsampling). In addition, focusing 8B ViT decoding on candidate windows reduces frame evaluations by **60.3%** (128 fixed frames vs. 322.8f average on raw video), offloading initial temporal search to the lightweight 1.04B scout.

4.3 Algorithmic Analysis: Window Planning and Windowing Topologies

1. Semantic Guidance vs. Random Windowing. As shown in Table 2, evaluating a random temporal window matching the average candidate duration ratio (40%) drops EtF1 to **3.61%** and tIoU to **24.18%**. This shows that temporal windowing without semantic guidance degrades grounding performance.

Table 2. Ablation of temporal window planning and windowing strategies on OMTG Bench ($N = 320$, 128-frame budget). Isolating semantic scouting, single-window bounding, and soft continuity merging.

Window Strategy	Planning Mode	C-Acc↑	tF1@.5	tIoU↑	EtF1↑	Err↓
(a) Random Temporal Crop	Random window (40% dur)	18.44%	24.29%	24.18%	3.61%	2.02
(b) Single-Window Baseline	Single bounding window	28.44%	38.26%	36.81%	16.86%	1.94
(c) Multi-Window (No Merge)	Split on gaps ($\rho = 1.0$)	28.44%	38.49%	37.12%	16.81%	1.94
(d) **Adaptive SGDE-128 (Full)**	**Adaptive window + merge** ($\rho = 0.7$)	**33.12%**	**41.82%**	**44.76%**	**19.14%**	**1.75**

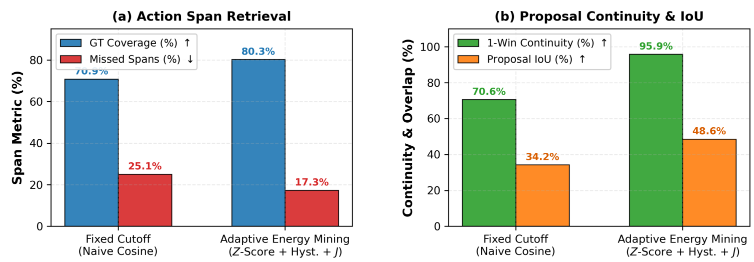

Fig. 4. Evaluation of timeline scoring and candidate extraction methods on OMTG Bench. (a) Action span retrieval: Z-score normalization and energy scoring increase Ground-Truth coverage to 80.3% and cut missed spans by 31.1% relative. (b) Proposal continuity and overlap: achieving 95.9% 1-window continuity and 48.6% proposal IoU.

2. Single-Window vs. Multi-Window Topology. Moving from raw video to a single continuous window (strategy b) improves Count Accuracy from 16.88% → **28.44%** and EtF1 from 12.29% → **16.86%**. However, forcing distant multi-event clusters into one window retains background frames between events, limiting tIoU to 36.81%. While splitting intervals without soft merging (strategy c) can fragment adjacent actions (16.81% EtF1), our proposed **Adaptive SGDE** with 70% soft continuity merging (strategy d) groups proximate clusters while isolating distant intervals, improving tIoU to **44.76%**, Count Accuracy to **33.12%**, and EtF1 to **19.14%**.

3. Timeline Scoring Function Evaluation. As shown in Fig. 4, raw cosine similarity with a fixed cutoff is sensitive to baseline shifts across videos, resulting in 25.1% missed action spans. Video-adaptive Z-score normalization with dual-threshold search ($\tau_{\text{high}} = 0.80, \tau_{\text{low}} = 0.25$) and energy scoring $\bar{J}(a, b)$ improves ground-truth coverage from 70.9% → **80.3%** (+9.4%), improves proposal IoU from 34.2% → **48.6%**, and reduces missed action spans by 31.1% relative.

4. Soft Merge Span Ratio Optimization (ρ). Fig. 5 shows the trade-off across merge ratio thresholds $\rho \in [30\%, 100\%]$. Low thresholds ($\rho \leq 40\%$) preserve all videos in 1-window prompts but prune fewer background frames (33.3%). High thresholds ($\rho \geq 90\%$) or gap-only merging split 7.5% − 14.7% of videos across multiple disjoint prompts, disrupting multi-event context. Setting $\rho = 70\%$

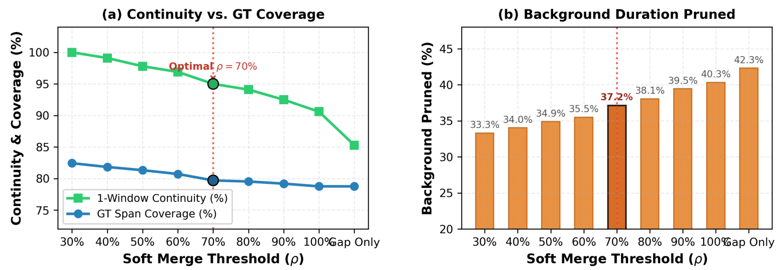

Fig. 5. Merge threshold (ρ) sweep on OMTG Bench. (a) Continuity vs. GT coverage: setting $\rho = 70\%$ retains 95.0% of videos within a single continuous prompt while covering 79.71% of GT action spans. (b) Background duration pruned: achieving 37.17% background pruning across the benchmark (up to 85.2% on videos $>$ 180s).

achieves a balanced trade-off: keeping **95.0% of videos in a single continuous prompt** while capturing **79.71% of GT action spans** and removing **37.17% of background frames** overall (up to 85.2% on videos $>$ 180s).

5 Conclusion

In this paper, we addressed the multi-event temporal grounding problem in Multimodal Large Language Models (OMTG). We showed that baseline performance drops stem from two main sources: prompt format misalignment and uninformative background frames in long videos. To address both challenges without fine-tuning, we presented **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**, a four-stage training-free framework combining 1.0 FPS temporal scouting, video-adaptive Z-score normalization, hysteresis energy mining, duration-adaptive windowing, and structured JSON decoding. On the complete OMTG benchmark ($N = 320$), Adaptive SGDE achieves strong performance (33.12% Count Accuracy, 19.86% EtF1) while reducing 8B Vision Transformer frame load by 60.3% (-54.0% net compute reduction).

Limitations and Future Work. The 1.0 FPS scouting rate establishes an 80.3% candidate coverage ceiling, filtering brief sub-second actions. Future work will investigate adaptive scout sampling rates and in-ViT motion-residual token pruning for scaling to hour-long video comprehension.

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